Robust Multi-Asset VaR and Expected Shortfall Forecasting via GJR-GARCH, Extreme Value Theory, and Vine Copula Models: Evidence from Indian Equity Markets

N. Devendra (ME21B127)
Indian Institute of Technology Madras
Under Prof. Neelesh Upadhye, Department of Mathematics, IIT Madras

1 Abstract

This paper develops and empirically evaluates a fallback-aware framework for forecasting portfolio Value-at-Risk (VaR) and Expected Shortfall (ES) in a rolling multi-asset setting, with application to Indian equity markets. The main focus is on the practical problem of Extreme Value Theory (EVT) threshold selection instability, where valid thresholds may not exist in all rolling windows, making fallback mechanisms necessary.

To address this, the framework combines EVT-based tail modeling with clearly defined fallback approaches, including a KDE-based method and a model-implied (no-KDE) alternative, so that the model remains usable even when EVT fails. Dependence across assets is modeled using vine copulas, and the framework is applied in a rolling-window setup across different portfolio sizes and forecast horizons.

Model performance is evaluated using the mean Fissler–Ziegel (FZ0) joint VaR–ES loss as the main criterion, with standard backtests such as Kupiec, Christoffersen, Dynamic Quantile, and Acerbi–Székely used as supporting checks. The results show that no single model performs best in all cases. Instead, performance depends on the forecast horizon and the number of assets, and the differences in FZ0 values are small(with maximum differences of approximately 0.037) .

These results suggest that more complex tail models do not always outperform simpler baseline models, which remain competitive in several cases. At the same time, the results show that fallback mechanisms are necessary in rolling EVT-based systems because valid thresholds cannot always be obtained. While no single fallback approach consistently dominates, the way fallback is designed and implemented can influence model performance in the subset of windows where EVT threshold selection fails.

Keywords: Value-at-Risk; Expected Shortfall; Extreme Value Theory; Fallback Modeling; Vine Copula; Fissler–Ziegel Loss; Indian Equity Markets

2 Introduction

Accurate measurement of market risk remains a central challenge in quantitative finance. Value-at-Risk (VaR) and Expected Shortfall (ES) are the primary regulatory risk measures under the Basel III and IV frameworks, yet their practical estimation for multi-asset equity portfolios is complicated by features such as volatility clustering, heavy tails, and asymmetric dependence. These characteristics are not adequately captured by standard normal-distribution assumptions in traditional models, leading to potential underestimation of extreme risks.

To address these limitations, a substantial body of literature combines GARCH-class volatility models with Extreme Value Theory (EVT) for tail estimation and copula models for capturing multivariate dependence (McNeil and Frey, 2000; Embrechts et al., 1997). However, many applied studies focus on developed markets, and EVT threshold selection is often handled using fixed or heuristic rules rather than being studied as a rolling-window problem.

In practice, EVT-based modeling presents an additional challenge: the threshold required for fitting the Generalized Pareto Distribution (GPD) must be re-selected in each rolling window, and its reliability may vary over time. In some windows, no statistically valid threshold satisfies standard goodness-of-fit, minimum exceedance, and stability criteria. This raises a practical question: how should tail risk be modeled in windows where EVT threshold selection fails? Despite its importance for real-world risk systems, this issue remains relatively underexplored in the rolling-window risk literature. Moreover, extending such frameworks to emerging markets, incorporating multi-day horizons, and conducting evaluation under overlapping data settings introduce additional complexities.

This paper addresses these issues through an empirical study using a portfolio constructed from stocks listed on the National Stock Exchange of India (NSE), including SBI, Infosys, HUL, and Tata Motors. The key contributions are as follows:

1. We explicitly address EVT threshold failure in a rolling-window setting. Instead of assuming that a valid threshold always exists, the framework identifies windows where threshold selection fails and handles them directly.
2. We define and compare fallback approaches within a multi-asset VaR/ES framework, including a KDE-based method and a model-implied (no-KDE) alternative, and study their impact on risk forecasts.
3. We adopt an overlap-aware evaluation framework for multi-day ($horizon = 10$) forecasts, where independence-based backtests are applied only to non-overlapping observations, ensuring valid statistical inference.
4. We provide empirical evidence on Indian equity portfolios across multiple configurations, using the mean Fissler–Ziegel (FZ0) joint loss as the primary evaluation metric, supported by standard backtesting procedures.

The remainder of the paper is organized as follows. Section 3 reviews the relevant literature. Section 4 describes the data. Section 5 presents the methodology. Section 6 reports the empirical results. Section 7 provides a discussion of the findings. Section 8 discusses managerial implications, and Section 9 concludes.

3 Literature

The motivation for advanced Value-at-Risk (VaR) modeling arises from the limitations of normality-based approaches. Early studies by Mandelbrot (1963) and Fama (1965) showed that financial return distributions are heavy-tailed, leading to underestimation of extreme losses under standard models. To account for time-varying volatility, Engle (1982) and Bollerslev (1986) introduced ARCH and GARCH models, while Glosten et al. (1993) extended this framework through the GJR-GARCH model to capture asymmetric responses to shocks. These models are widely used as the filtering step in risk modeling.

A common approach for modeling tail risk combines GARCH filtering with Extreme Value Theory (EVT) using the Peaks-over-Threshold (POT) method (McNeil and Frey, 2000; Embrechts et al., 1997). EVT provides a theoretically justified framework for modeling extremes, with the Generalized Pareto Distribution (GPD) established as the limiting distribution for exceedances above a high threshold (Balkema and de Haan, 1974; Pickands, 1975). However, the practical performance of EVT depends heavily on threshold selection. Issues related to threshold instability in rolling settings are not always explicitly addressed in applied risk models, with many studies relying on fixed quantile-based thresholds such as the 90th, 95th, or 99th percentile. While convenient, such thresholds may not remain statistically valid across rolling windows as tail behavior changes over time.

In multivariate settings, dependence across assets is modeled using copulas. Sklar's theorem (Sklar, 1959) provides the foundation for separating marginal distributions from dependence structures. Vine copula constructions (Bedford and Cooke, 2001; Czado, 2019) extend these ideas to higher dimensions and are commonly used in multi-asset portfolio risk modeling.

The evaluation of risk models has also received considerable attention. Kupiec (1995) and Christoffersen (1998) developed likelihood-based tests for VaR exceedances and their independence. Engle and Manganelli (2004) proposed the Dynamic Quantile (DQ) test for detecting dynamic misspecification. For Expected Shortfall, Acerbi and Székely (2014) introduced simulation-based testing methods. More recently, Fissler and Ziegel (2016) showed that VaR and ES are jointly elicitable, enabling the use of consistent scoring functions such as the FZ0 loss. In multi-day forecasting settings, overlapping observations introduce dependence, which requires careful interpretation of independence-based backtests.This study builds on the existing literature by examining EVT-based modeling within a rolling multi-asset framework, with particular attention to the role of fallback mechanisms.

4 Data

4.1 Assets and Sample Period

The study uses daily closing prices sourced from Yahoo Finance for four assets listed on the National Stock Exchange of India: State Bank of India (SBIN.NS), Infosys Limited (INFY.NS), Hindustan Unilever Limited (HINDUNILVR.NS), and Tata Motors Limited (TATAMOTORS.NS). These assets represent distinct economic sectors—banking, information technology, consumer staples, and automotive manufacturing—providing diversification in the four-asset portfolio setting. The two-asset configuration uses SBIN.NS and INFY.NS.

The full data span covers April 27, 2011 to May 27, 2025, yielding approximately 3,472 trading days. Log returns are computed as $r_t = 100 \times \ln(P_t/P_{t-1})$, where P_t denotes the closing price at time t and reported as percentage log-returns. The first 1,000 observations form the initial estimation window, after which a rolling-window approach is employed, with the model re-estimated in each window and subsequent observations used for out-of-sample evaluation.

4.2 Tail Behavior Diagnostics: Evidence of Tail Mismatch

Before presenting the estimation framework, we examine the tail behavior of the standardized residuals to assess whether the fitted parametric model adequately captures extreme risk. Tail diagnostics are based on Q–Q plots and Hill estimates computed from an AR(1)-GJR-GARCH(1,1) fit with Student-t assumption over the initial estimation window. The Hill estimator (Hill, 1975) provides a measure of tail heaviness through the extreme value index γ, defined as:

$$\hat{\gamma}_k = \frac{1}{k} \sum_{i=1}^{k} \left[\log X_{(n-i+1)} - \log X_{(n-k)} \right] \tag{1}$$

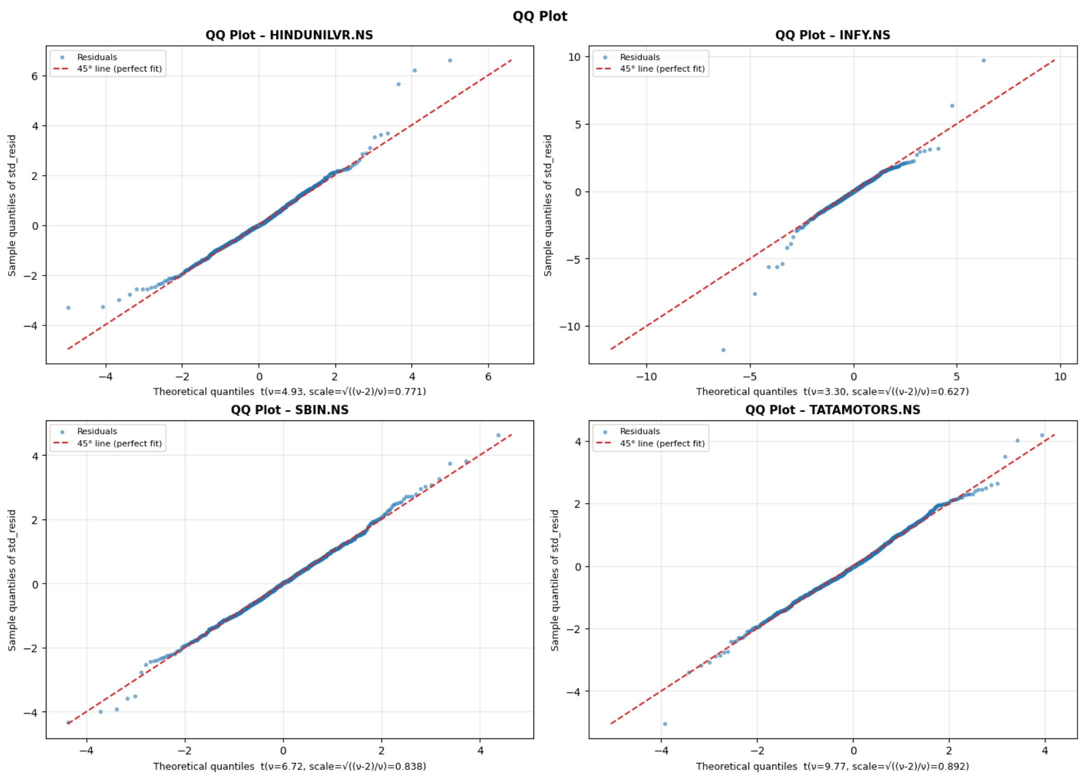
where k denotes the number of upper order statistics (extreme observations) used in the estimation, with $1 \leq k < n$; n is the sample size; $X_{(i)}$ represents the i-th order statistic of the sample sorted in ascending order, i.e., $X_{(1)} \leq \cdots \leq X_{(n)}$; $X_{(n-i+1)}$ is the i-th largest observation; $X_{(n-k)}$ is the $(k+1)$-th largest observation and acts as a data-dependent threshold. The estimator computes the average log-excess of the largest k observations over the threshold $X_{(n-k)}$, providing a measure of tail heaviness.

Table 1 summarizes the Hill estimates for both upper and lower tails across all assets. The estimated values range approximately from 0.32 to 0.47, indicating substantial tail heaviness in this window. In comparison, the fitted Student-t distribution implies a tail index of $\gamma = 1/\nu$, which lies between approximately 0.10 and 0.30 depending on the asset. In this initial window, the empirical Hill estimates are consistently higher than the model-implied values, suggesting heavier tails than those captured by the fitted GJR-GARCH model with Student-t innovations.

Asset	Hill γ (Upper)	Hill γ (Lower)	$1/\nu$	Upper Tail vs t	Lower Tail vs t
HINDUNILVR.NS	0.3583	0.3873	0.2027	Heavier	Heavier
INFY.NS	0.4127	0.4730	0.3032	Heavier	Heavier
SBIN.NS	0.3650	0.3611	0.1488	Heavier	Heavier
TATAMOTORS.NS	0.3217	0.3258	0.1024	Heavier	Heavier

Table 1: Hill Tail Index Estimates vs. Student-t (Initial Estimation Window)

These findings are supported by Q–Q plot diagnostics, which show that while the central region of the distribution is reasonably well captured, noticeable deviations occur in the extreme quantiles. In particular, the empirical tails diverge from those implied by the fitted Student-t distribution.



It is important to note that these diagnostics are based on a single estimation window and do not imply that the same tail behavior persists throughout the entire sample. Tail properties may vary across rolling windows, including periods of comparatively lighter tails. Therefore, these results should be interpreted as motivation for adaptive tail modeling.

In particular, the evidence suggests that a fixed parametric specification may not adequately capture tail behavior across all periods. This motivates the use of modeling approaches that can adapt to changing tail conditions over time, such as EVT-based methods with rolling threshold selection and fallback mechanisms when standard EVT assumptions are not satisfied.

5 Methodology

For each rolling window, the following steps are executed in sequence: (i) AR(1)–GJR-GARCH(1,1) fitting, (ii) EVT threshold selection, (iii) fallback-aware hybrid CDF construction and Probability Integral Transform (PIT), (iv) vine copula fitting, and (v) Monte Carlo simulation for VaR and ES computation.

5.1 Return Model: AR(1)–GJR-GARCH(1,1)

For each asset i and each rolling window, returns r_t are modeled as:

$$r_t = \mu_t + \varepsilon_t, \quad \mu_t = c + \phi r_{t-1} \tag{2}$$

$$\varepsilon_t = \sigma_t z_t, \quad z_t \sim t(\nu) \tag{3}$$

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \gamma\varepsilon_{t-1}^2 \mathbf{1}_{\{\varepsilon_{t-1}<0\}} + \beta\sigma_{t-1}^2 \tag{4}$$

where r_t denotes the return at time t and they are defined such that losses correspond to negative values, μ_t is the conditional mean, ε_t is the innovation, and σ_t^2 is the conditional variance. The innovations z_t follow a Student-t distribution with ν degrees of freedom.

The leverage term $\gamma\varepsilon_{t-1}^2 \mathbf{1}_{\{\varepsilon_{t-1}<0\}}$ captures the asymmetric impact of negative shocks on volatility. Parameters $\{c, \phi, \omega, \alpha, \gamma, \beta, \nu\}$ are estimated by maximum likelihood within each rolling window.Standardized residuals are computed as $\hat{z}_t = \varepsilon_t/\sigma_t$ and used for tail modeling.

As a benchmark, we consider a Gaussian copula AR(1)–GJR-GARCH(1,1) model with the same mean and variance specification, but with normally distributed innovations and no EVT-based tail modeling.

5.2 EVT Threshold Selection

Threshold selection is a key step in Peaks-over-Threshold (POT) modeling. We implement an automated multi-criterion procedure separately for the lower and upper tails of standardized residuals.

Threshold search: For each tail, 50 candidate thresholds are evaluated over the range corresponding to the 80th to 99.5th percentiles of the absolute residuals.

For each candidate threshold u:

- Exceedances above u are fitted using a Generalized Pareto Distribution (GPD) with parameters (ξ, σ) estimated by maximum likelihood.

Acceptance criteria: A threshold is accepted if it satisfies all of the following:

- **Minimum exceedances:** At least 100 exceedances.
- **Kolmogorov–Smirnov test:** p-value > 0.05.
- **Anderson–Darling test:** Bootstrap p-value > 0.05 (300 simulations).
- **Parameter stability:** Estimated GPD parameters remain stable in a neighbourhood of the threshold.

Threshold selection outcome:

- **Full-pass case:** One or more thresholds satisfy all criteria. The threshold with the largest number of exceedances is selected and EVT is applied.
- **Sparse case:** A threshold satisfies goodness-of-fit and parameter stability criteria but fails the minimum exceedance requirement. In this case, EVT is not considered reliable, and fallback modeling is triggered.
- **Invalid case:** No threshold satisfies both the goodness-of-fit and parameter stability criteria. EVT is not applied, and fallback modeling is triggered.

5.3 Fallback Regimes

Fallback modeling is a central component of the framework, as EVT threshold selection does not succeed in all rolling windows. Based on the threshold selection procedure, each tail in each window is classified into full-pass, sparse, and invalid cases. Fallback is applied in sparse and invalid cases, while EVT is used only in full-pass cases.

When fallback is required, two alternative approaches are evaluated:

- **No-KDE fallback (GJR-GARCH implied tails):** The tail is modeled using the parametric distribution implied by the AR(1)–GJR-GARCH(1,1) model fitted in the first step. In sparse windows, the candidate threshold with the largest exceedance count is used to separate the tail from the central region. In failure windows, the threshold with the maximum exceedances within the search range (80th–99.5th percentile) is selected. KDE is applied only to the central region.
- **KDE fallback:** When EVT thresholding is not reliable, KDE is applied to the entire distribution, including both the central region and the affected tail, avoiding the use of a poorly fitted GPD.

The with-KDE and no-KDE variants differ only in their fallback treatment. In full-pass windows, both approaches are identical.

5.4 Hybrid CDF and Probability Integral Transform

For each asset, a hybrid semi-parametric cumulative distribution function (CDF) is constructed by combining EVT-based GPD tails with a kernel density estimate (KDE) for the central region. The exact form of the hybrid CDF depends on whether EVT threshold selection succeeds in the lower and upper tails and on the fallback design adopted when EVT fitting is unreliable.

KDE Fallback Design

Case 1: EVT succeeds in both tails

$$F(x) = \frac{n_l}{n}\left(1 - G_L(u_L - x)\right), \qquad x < u_L, \tag{5}$$

$$F(x) = \frac{n_l}{n} + \frac{n_c}{n}\hat{F}_{\mathrm{KDE}}(x), \qquad u_L \le x \le u_U, \tag{6}$$

$$F(x) = 1 - \frac{n_u}{n}\left(1 - G_U(x - u_U)\right), \qquad x > u_U. \tag{7}$$

Case 2: Lower-tail fallback, upper-tail EVT

$$F(x) = \left(1 - \frac{n_u}{n}\right)\hat{F}_{\mathrm{KDE}}(x), \qquad x \le u_U, \tag{8}$$

$$F(x) = 1 - \frac{n_u}{n}\left(1 - G_U(x - u_U)\right), \qquad x > u_U. \tag{9}$$

Case 3: Lower-tail EVT, upper-tail fallback

$$F(x) = \frac{n_l}{n}\left(1 - G_L(u_L - x)\right), \qquad x < u_L, \tag{10}$$

$$F(x) = \frac{n_l}{n} + \left(1 - \frac{n_l}{n}\right)\hat{F}_{\mathrm{KDE}}(x), \qquad x \ge u_L. \tag{11}$$

Case 4: Fallback in both tails

$$F(x) = \hat{F}_{\mathrm{KDE}}(x). \tag{12}$$

No-KDE Fallback Design

Case 1: EVT succeeds in both tails

Identical to the full-pass specification given previously.

Case 2: Lower-tail fallback, upper-tail EVT

$$F(x) = \frac{n_l}{n}F_{\mathrm{GARCH}}(x), \qquad x < u_L, \tag{13}$$

$$F(x) = \frac{n_l}{n} + \frac{n_c}{n}\hat{F}_{\mathrm{KDE}}(x), \qquad u_L \le x \le u_U, \tag{14}$$

$$F(x) = 1 - \frac{n_u}{n}\left(1 - G_U(x - u_U)\right), \qquad x > u_U. \tag{15}$$

Case 3: Lower-tail EVT, upper-tail fallback

$$F(x) = \frac{n_l}{n}\left(1 - G_L(u_L - x)\right), \qquad x < u_L, \tag{16}$$

$$F(x) = \frac{n_l}{n} + \frac{n_c}{n}\hat{F}_{\mathrm{KDE}}(x), \qquad u_L \le x \le u_U, \tag{17}$$

$$F(x) = 1 - \frac{n_u}{n} + \frac{n_u}{n}F_{\mathrm{GARCH}}(x), \qquad x > u_U. \tag{18}$$

Case 4: Fallback in both tails

$$F(x) = \frac{n_l}{n}F_{\mathrm{GARCH}}(x), \qquad x < u_L, \tag{19}$$

$$F(x) = \frac{n_l}{n} + \frac{n_c}{n}\hat{F}_{\mathrm{KDE}}(x), \qquad u_L \le x \le u_U, \tag{20}$$

$$F(x) = 1 - \frac{n_u}{n} + \frac{n_u}{n}F_{\mathrm{GARCH}}(x), \qquad x > u_U. \tag{21}$$

In the preceding expressions, u_L and u_U denote the lower and upper thresholds, respectively; n is the total number of observations; and n_l, n_c, and n_u are the numbers of observations assigned to the lower tail, central region, and upper tail, respectively, with $n = n_l + n_c + n_u$. The functions $G_L(\cdot)$ and $G_U(\cdot)$ denote the fitted Generalized Pareto Distribution (GPD) cumulative distribution functions for the lower and upper tails, applied to the exceedances $u_L - x$ and $x - u_U$, respectively. $\hat{F}_{\mathrm{KDE}}(x)$ denotes the kernel density estimate (KDE)-based cumulative distribution function, appropriately normalized over the region over which KDE is applied. Finally, $F_{\mathrm{GARCH}}(x)$ denotes the normalised parametric cumulative distribution function implied by the fitted AR(1)–GJR-GARCH(1,1) model with Student-t innovations.

Probability Integral Transform (PIT) values are computed for each asset from $F(x)$, transforming standardized residuals to uniform marginals on $[0, 1]$. Correct specification of the hybrid CDF implies that the PIT values should be uniform; we verify this via KS uniformity tests at each window as a diagnostic and those results are satisfactory.

5.5 Vine Copula Selection and Fitting

Multivariate dependence among the PIT-transformed marginals is modeled using vine copulas (pair-copula constructions), considering Student-t and BB7 bivariate families. For two-asset portfolios, a single bivariate copula is fitted. For four-asset portfolios, the vine structure is determined automatically from the data.

Copula selection is carried out using a three-stage procedure combining goodness-of-fit, tail dependence consistency, and a rolling cross-validation criterion.

Stage 1 — Goodness-of-Fit: The Rosenblatt transformation-based Cramér–von Mises (CvM) goodness-of-fit test (Genest et al., 2009) is applied. For each candidate copula, 10,000 observations are simulated, and the test is repeated 1,000 times with varying random seeds. A copula is accepted if the resulting p-value exceeds 0.05.

Stage 2 — Tail Dependence Consistency: The empirical lower tail dependence λ_L is estimated from the PIT-transformed data. The model-implied tail dependence λ_{model} is obtained via Monte Carlo simulation. A copula is accepted if: (i) the empirical λ_L lies within the 95% bootstrap confidence interval of the model (based on 1,000 replications of 500 simulations each), and (ii) $|\lambda_{\mathrm{model}} - \lambda_{\mathrm{emp}}| \leq 0.05$.

Stage 3 — Rolling Cross-Validation (FZ0-based): Candidate copulas that pass the first two stages are compared using a rolling cross-validation rule based on recent forecasting performance. Specifically, for each candidate copula, the mean Fissler–Ziegel (FZ0) loss is computed over the last $L = 50$ rolling windows, using realized returns and corresponding VaR and ES forecasts.

For initial windows where fewer than L past observations are available, the Akaike Information Criterion (AIC) is used as a fallback selection criterion.

Moderate variations in L (e.g., $L = 25$,$L = 50$, $L = 100$) do not materially affect the final mean FZ0 loss of model much(Difference between L=25 ,L=50 and L=50, L=100 has maximum value of around 0.006), which can be seen in Table 2, suggesting that the results are not highly sensitive to the specific choice of L.So finally, we considered L as 50 for evaluation.

Table 2: Sensitivity of Mean FZ0 Loss to the Choice of Rolling Selection Window L

Horizon	Setting	$L=25$			$L=50$			$L=100$		
		Baseline	No-KDE	KDE	Baseline	No-KDE	KDE	Baseline	No-KDE	KDE
H=1	S=1, 2A	1.1293	1.1307	1.1299	1.1293	1.1318	1.1307	1.1293	1.1314	1.1304
H=1	S=10, 2A	0.9110	0.9327	0.9322	0.9110	0.9334	0.9331	0.9110	0.9367	0.9361
H=1	S=1, 4A	1.0197	1.0170	1.0168	1.0197	1.0170	1.0177	1.0197	1.0144	1.0150
H=1	S=10, 4A	1.0232	1.0161	1.0170	1.0232	1.0226	1.0233	1.0232	1.0280	1.0285
H=10	S=1, 2A	2.2688	2.2668	2.2642	2.2688	2.2670	2.2643	2.2688	2.267	2.2649
H=10	S=10, 2A	2.3056	2.2761	2.2688	2.3056	2.2758	2.2686	2.3056	2.2767	2.2694
H=10	S=1, 4A	2.2439	2.2577	2.2569	2.2439	2.2587	2.2577	2.2439	2.2602	2.2588
H=10	S=10, 4A	2.26387	2.2708	2.2708	2.2639	2.2715	2.2720	2.2639	2.2735	2.2735

Notes: H denote the forecast horizon and S the refit step size.

5.6 VaR and ES Computation via Monte Carlo Simulation

Let h denote the forecast horizon and s the refit step size. We consider $h \in \{1, 10\}$ and $s \in \{1, 10\}$.

When $s = 1$, forecasts are generated at every time point, resulting in overlapping observations for $h > 1$. When $s >= h$, forecasts are generated at non-overlapping intervals.

For each rolling window, Value-at-Risk (VaR) and Expected Shortfall (ES) at confidence level $\alpha = 5\%$ are computed for an equal-weighted portfolio using Monte Carlo simulation.

One-day horizon ($h = 1$): 10,000 joint residuals are simulated from the selected vine copula and transformed into asset returns using the inverse hybrid CDF along with current conditional mean and volatility. VaR is defined as the α-quantile (5th percentile) of simulated portfolio returns, and ES as the average of returns below the VaR threshold.

Ten-day horizon ($h = 10$): Simulation is performed sequentially. At each step, residuals are simulated, transformed into returns, and the GJR-GARCH recursion is used to update conditional mean and variance. The cumulative 10-day return is obtained by summing simulated returns across the horizon. This approach captures the propagation of volatility dynamics over time.

5.7 Evaluation Framework

5.7.1 Primary Criterion: Mean FZ0 Joint Loss

The primary model comparison criterion is the mean FZ0 joint VaR–ES loss of Fissler and Ziegel (2016):

$$\mathrm{FZ0}(\mathrm{VaR}, \mathrm{ES}; y_t) = \frac{1}{\alpha ES_t^\alpha} I_t^\alpha (y_t - \mathrm{VaR}_t^\alpha) + \frac{\mathrm{VaR}_t^\alpha}{ES_t^\alpha} + \log(-ES_t^\alpha) - 1 \tag{22}$$

where $I_t^\alpha = \mathbf{1}\{y_t \le \mathrm{VaR}_t^\alpha\}$ is the indicator function, which takes value 1 if the realized portfolio return falls below the predicted VaR (i.e., a violation occurs), and 0 otherwise. Here losses are represented as negative returns, accordingly VaR and ES will be negative. A lower FZ0 loss indicates superior joint calibration of both VaR and ES. Mean FZ0 is the primary criterion. The FZ0 comparison is valid across all eight experimental configurations since it is computed observation-by-observation and does not require non-overlapping windows.

5.7.2 Secondary Criteria: Formal Backtests

Secondary evaluation uses the following tests:

- **Kupiec (1995) likelihood-ratio test:** tests whether the empirical violation rate matches the nominal $\alpha = 5\%$.
- **Christoffersen (1998) independence test:** tests whether VaR violations are serially independent.
- **Dynamic Quantile test** (Engle and Manganelli, 2004) with lags 1 and 4: tests for absence of autocorrelation in the hit sequence.
- **Acerbi–Székely (2014) ES test:** the null hypothesis that the ES model is correctly specified is tested by generating 10,000 null datasets using a global seed of 42.

For $h = 10$, step-size-1 configurations produce 2,462 overlapping observations. The overlap implies that consecutive forecasts share a large portion of underlying data, inducing strong serial dependence in the violation sequence. As a result, the independence assumption underlying the Christoffersen and DQ tests is violated, rendering these tests invalid as formal hypothesis tests; they are therefore reported as diagnostic measures only. The Kupiec test is less sensitive to dependence but may suffer from reduced effective sample size, leading to distorted p-values. The Acerbi–Székely test relies on a Monte Carlo null distribution generated under an i.i.d. assumption; in the presence of overlap-induced dependence, this may result in size distortions, and the test is therefore interpreted with caution. For $h = 10$, step-size-10, approximately 247 non-overlapping observations are available and all four tests are fully applicable. For $h = 1$ in both step sizes there will be only non-overlapping observations, so all tests are valid.

6 Empirical Results

This section presents the empirical results in four steps. First, we compare baseline and EVT-based models using the FZ0 loss as the primary ranking criterion. Second, we examine fallback frequency . Third, we review formal backtesting results as supporting evidence. Finally, we discuss overlapping multi-day results, where independence-based tests must be interpreted with caution.

6.1 Model Comparison Based on FZ0

Table 3 reports results across all eight configurations. The primary comparison is based on mean FZ0 loss, with lower values indicating better joint VaR–ES performance. Backtests are reported as supporting evidence.

Differences in mean FZ0 between models are small (approximate max value is 0.037). No single model uniformly dominates in all configurations. Instead, model performance varies across horizons and portfolio sizes.

At the one-day horizon ($h = 1$), the Gaussian baseline performs best for the two-asset portfolio under both step sizes, while the no-KDE variant achieves the lowest FZ0 for the four-asset portfolio.

At the ten-day horizon ($h = 10$), the KDE variant performs best for the two-asset portfolio, whereas the baseline model performs best for the four-asset portfolio under both step sizes.

These results indicate that more complex tail modeling does not consistently outperform the simpler baseline. There is no single consistently dominant model. Instead, the ranking varies across configurations, with different models achieving the lowest FZ0 in different settings. This pattern suggests that model performance is context-dependent rather than driven by a uniform advantage of any one specification.

Table 3: Full Backtesting Results Across All Eight Experimental Configurations

Horizon	Setting	Model	Viol. Rate	Kupiec p	Christoffersen p	DQ p (lag 1)	DQ p (lag 4)	Acerbi-Székely p	FZ0 Mean
H=1	S=1, 2A	Baseline	4.29%	0.097	0.784	0.263	0.667	0.511	1.1293
		No-KDE	5.02%	0.967	0.924	0.994	0.913	0.654	1.1318
		KDE	4.94%	0.886	0.991	0.990	0.955	0.620	1.1307
H=1	S=10, 2A	Baseline	3.63%	0.299	0.409	0.544	0.149	0.217	0.9110
		No-KDE	4.84%	0.907	0.268	0.727	0.389	0.366	0.9334
		KDE	4.84%	0.907	0.268	0.727	0.389	0.365	0.9331
H=1	S=1, 4A	Baseline	4.45%	0.203	0.348	0.296	0.263	0.622	1.0197
		No-KDE	4.82%	0.673	0.345	0.569	0.428	0.534	1.0170
		KDE	4.78%	0.606	0.323	0.521	0.404	0.514	1.0177
H=1	S=10, 4A	Baseline	3.23%	0.172	0.464	0.408	0.798	0.376	1.0232
		No-KDE	3.63%	0.299	0.409	0.544	0.836	0.382	1.0226
		KDE	3.63%	0.299	0.409	0.544	0.836	0.381	1.0233
H=10	S=1, 2A†	Baseline	3.86%	0.007‡	0.000*	0.000*	0.000*	0.147	2.2688
		No-KDE	4.39%	0.154	0.000*	0.000*	0.000*	0.438	2.2670
		KDE	4.31%	0.106	0.000*	0.000*	0.000*	0.397	2.2643
H=10	S=10, 2A	Baseline	3.64%	0.305	0.320	0.363	0.616	0.405	2.3056
		No-KDE	4.05%	0.479	0.406	0.542	0.644	0.471	2.2758
		KDE	4.05%	0.479	0.406	0.542	0.644	0.464	2.2686
H=10	S=1, 4A†	Baseline	5.00%	0.993	0.000*	0.000*	0.000*	0.910	2.2439
		No-KDE	5.52%	0.240	0.000*	0.000*	0.000*	0.977	2.2587
		KDE	5.40%	0.366	0.000*	0.000*	0.000*	0.966	2.2577
H=10	S=10, 4A	Baseline	4.45%	0.688	0.451	0.587	0.693	0.642	2.2639
		No-KDE	5.67%	0.637	0.760	0.928	0.648	0.777	2.2715
		KDE	5.67%	0.637	0.760	0.928	0.648	0.776	2.2720

*H = horizon; S = step size; A = number of assets; Viol. Rate = violation rate (target: 5%); FZ0 Mean = mean Fissler–Ziegel joint loss . † = independence tests are diagnostic only due to overlap. Christoffersen p and DQ p marked with * are structurally zero due to serially dependent overlapping observations ($h = 10$, S=1); treat as diagnostic only. ‡ denotes Kupiec rejection at 5% for the baseline in H=10, S=1, 2A configuration.*

6.2 Fallback Frequency and EVT Stability

Table 4 summarizes EVT threshold selection outcomes across assets and configurations. Fallback is triggered when EVT threshold selection fails or is weakly supported, and therefore represents a key component of the modeling framework.

Fallback occurs in a non-trivial fraction of windows, ranging from 0% to 16.35%, confirming that EVT threshold failure is a recurring feature in rolling estimation rather than an isolated event.So handling fallback is important.There is substantial heterogeneity across assets and between upper and lower tails. In some cases, fallback is driven primarily by invalid thresholds, indicating failure of the GPD fit, while in others it is driven by sparse exceedances, indicating insufficient tail data.These patterns imply that EVT applicability varies across assets, tails, and time, rather than being uniformly reliable.

The EVT-based models (with-KDE and no-KDE) differ from the Gaussian baseline through a broader modeling structure, including more flexible tail modeling and dependence representation. These differences persist even in windows where EVT threshold selection succeeds.

With-KDE and no-KDE variants differ only in windows where EVT cannot be reliably applied. The with-KDE model replaces the affected tail using a nonparametric KDE approach, while the no-KDE model relies on the parametric distribution implied by the GJR-GARCH specification. As a result, differences between these two models arise exclusively in fallback windows.Although fallback occurs in only a subset of windows, its impact is not negligible. In some configurations, the KDE-based approach achieves lower FZ0, while in others the no-KDE approach performs better. This indicates that the way fallback is handled can influence model performance, even if overall differences remain small.

Table 4: EVT Threshold Selection Outcomes by Asset and Configuration

Asset	Setting	Tail	Total Windows	Invalid	Sparse	No Fallback	Fallback Rate (%)
INFY.NS	S=1	Lower	2471	0	0	2471	0.0%
INFY.NS	S=1	Upper	2471	282	122	2067	16.35%
SBIN.NS	S=1	Lower	2471	26	322	2123	14.08%
SBIN.NS	S=1	Upper	2471	27	13	2431	1.62%
INFY.NS	S=10	Lower	248	0	0	248	0.0%
INFY.NS	S=10	Upper	248	26	14	208	16.13%
SBIN.NS	S=10	Lower	248	3	34	211	14.92%
SBIN.NS	S=10	Upper	248	4	1	243	2.02%
TATAMOTORS.NS	S=1	Lower	2471	61	157	2253	8.82%
TATAMOTORS.NS	S=1	Upper	2471	3	14	2454	0.69%
HINDUNILVR.NS	S=1	Lower	2471	25	53	2393	3.16%
HINDUNILVR.NS	S=1	Upper	2471	120	63	2288	7.41%
HINDUNILVR.NS	S=10	Lower	248	3	5	240	3.23%
HINDUNILVR.NS	S=10	Upper	248	14	4	230	7.26%
TATAMOTORS.NS	S=10	Lower	248	7	14	227	8.47%
TATAMOTORS.NS	S=10	Upper	248	0	1	247	0.40%

Notes: Fallback Rate = (invalid + sparse) / total × 100. Valid = no fallback triggered.

6.3 Backtesting Results (Supporting Evidence)

Backtesting results reported in Table 3 serve as supporting evidence for the FZ0-based comparison.

At the one-day horizon ($h = 1$), all models pass the standard backtests. Differences in model performance are therefore driven primarily by efficiency, as captured by FZ0, rather than by violations of statistical assumptions.

At the ten-day horizon under non-overlapping evaluation ($h = 10$, $s = 10$), all models again satisfy the backtesting criteria, confirming that VaR and ES forecasts are broadly well calibrated across models.

An exception arises in the overlapping setting ($h = 10$, $s = 1$), where the baseline model fails the Kupiec test in the two-asset configuration. However, this case must be interpreted with caution due to the presence of overlapping observations, which introduce dependence and can distort test statistics. In this setting, independence-based tests are not formally valid, and FZ0 remains the primary evaluation criterion.

Overall, backtests provide limited discrimination between models and indicate that all specifications achieve broadly adequate calibration. As a result, differences in performance are better captured by FZ0, where variations are small and depend on the specific configuration.

6.4 Overlapping Multi-Day Results ($h = 10$, $s = 1$)

The overlapping setting ($h = 10$, $s = 1$) produces a larger number of forecast observations, but these observations are strongly dependent due to shared underlying return paths. As a result, independence-based backtests such as Christoffersen and Dynamic Quantile yield degenerate outcomes, with p-values collapsing to zero across all models. These zero values are structural and arise from violation of the independence assumption rather than from model misspecification, and should therefore be interpreted as diagnostic only.

Other backtests must also be interpreted with caution in this setting. The Kupiec test, while not directly testing independence, can be affected by the reduced effective sample size under overlapping observations. Similarly, the Acerbi–Székely ES test relies on a simulation-based null distribution and may exhibit size distortions when the underlying observations are dependent. These tests are therefore reported for completeness but are not used for formal model comparison.

In this setting, FZ0 remains the primary evaluation metric. The FZ0 results show small differences across models. In the two-asset case, the KDE variant achieves the lowest FZ0, while in the four-asset case the baseline performs best.

7 Discussion

The empirical results highlight that EVT threshold failure is not a marginal issue in rolling risk estimation, but a recurring feature that must be explicitly addressed. Fallback is triggered in up to 16.35% of rolling windows, with substantial variation across assets and between upper and lower tails. This indicates that EVT-based tail modeling cannot be assumed to be uniformly applicable over time, and that any practical implementation must incorporate a mechanism to handle periods in which threshold selection is unreliable.

The role of fallback becomes particularly relevant when comparing the with-KDE and no-KDE variants within the EVT-based framework. These two models are identical in windows where EVT threshold selection succeeds, and therefore any differences between them arise only in fallback windows.The results show that neither fallback approach consistently dominates: in some configurations the KDE-based fallback achieves lower FZ0, while in some other cases the No KDE-based fallback performs better. This indicates that the way fallback is implemented can influence model performance , even though its overall effect remains limited due to the relatively small proportion of fallback occurrences.

Differences in FZ0 across models are consistently small, and no single model provides uniformly superior performance. While EVT-based models offer improvements in certain configurations, the Gaussian baseline remains competitive throughout. This indicates that performance depends on the interaction between horizon length, portfolio dimension, rather than on a single dominant modeling approach.

The overlapping multi-day setting further illustrates the importance of careful evaluation. The dependence induced by overlapping observations limits the interpretability of standard backtesting procedures, particularly those based on independence assumptions. In this context, FZ0 provides a more reliable basis for comparison, as it does not rely on independence.

Taken together, these findings suggest that the primary challenge in rolling EVT-based VaR and ES forecasting lies not only in modeling tail behavior, but also in ensuring robustness when EVT cannot be reliably applied. The fallback-aware framework addresses this issue directly, it provides a systematic and stable approach to handling threshold instability in practice.

8 Managerial Implications

The results highlight several practical considerations for the design and implementation of risk systems.

Fallback-aware risk systems: Risk systems based on EVT should incorporate an explicit fallback policy to ensure continuity of VaR and ES forecasts when threshold selection fails. The choice of fallback method can affect results in these periods, even if overall performance differences remain small.

Monitoring fallback as a model-health indicator: Since fallback is triggered in a measurable subset of windows and varies across assets and tails, tracking fallback frequency provides useful diagnostic information. High or persistent fallback rates may indicate instability in tail behavior or limitations of the EVT specification, and can serve as a model-health indicator in operational risk systems.

Overlap-aware evaluation for multi-day risk: For multi-day horizons ,in some cases overlapping observations introduce strong dependence across forecasts. Standard backtesting procedures based on independence assumptions can produce misleading results in this setting. Risk systems should therefore adopt overlap-aware evaluation protocols, distinguishing between diagnostic and formally valid tests, and relying more heavily on metrics such as FZ0 that do not require independence.

Backtests versus scoring-based evaluation: The results show that passing standard backtests does not necessarily imply superior model performance. All models considered achieve broadly acceptable calibration, yet differences remain in FZ0. This highlights that regulatory backtesting and scoring-rule-based evaluation serve different purposes: the former ensures adequacy, while the latter provides a more sensitive ranking of competing models.

9 Conclusions

This paper develops and evaluates a multi-horizon portfolio risk estimation framework combining AR(1)–GJR-GARCH filtering, automated EVT threshold selection, and vine copula dependence modeling in a rolling-window setting applied to Indian equity markets.

The empirical results highlight that EVT threshold failure is operationally relevant in practice. Across assets and tails, valid thresholds cannot be consistently identified in all rolling windows, with fallback triggered in up to 16.35% of cases. This indicates that EVT-based tail modeling cannot be assumed to apply uniformly over time, and that any practical implementation must explicitly account for threshold instability.

The comparison of models shows that no single specification uniformly dominates across all horizons and portfolio configurations. Differences in FZ0 are consistently small, and relative performance varies across settings. While EVT-based models provide improvements in some cases, the Gaussian baseline remains competitive throughout.

Within the EVT-based framework, the choice of fallback policy can influence results in the subset of windows where threshold selection is not reliable. The with-KDE and no-KDE approaches perform differently across configurations, with neither approach consistently outperforming the other. This indicates how we design fallback is important.

The results also emphasize the importance of appropriate evaluation protocols. For multi-day horizons, overlapping observations introduce dependence that invalidates standard independence-based backtests such as Christoffersen and Dynamic Quantile. In such settings, these tests should be treated as diagnostic, and evaluation should rely more heavily on measures such as FZ0 that remain valid under dependence.

Overall, the contribution of this study lies in explicitly addressing EVT threshold instability within a rolling multi-asset risk framework and demonstrating its implications for model design and evaluation. Future work may extend this analysis to larger asset universes, stress-period analysis, and application of machine learning techniques to further assess the robustness of fallback-aware risk modeling approaches.

References

Acerbi, C., and Székely, B. (2014). Back-testing expected shortfall. *Risk*, 27(11), 76–81.

Balkema, A.A., and de Haan, L. (1974). Residual life time at great age. *Annals of Probability*, 2(5), 792–804.

Bedford, T., and Cooke, R.M. (2001). Probability density decomposition for conditionally dependent random variables modeled by vines. *Annals of Mathematics and Artificial Intelligence*, 32, 245–268.

Bhatt, V., and Vipul. (2015). Forecasting volatility of Indian equity market: Comparison of GARCH models. *International Journal of Finance and Economics*, 20(2), 190–208.

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.

Buch-Kromann, T., et al. (2009). Nonparametric estimation of the loss distribution. *Working Paper*.

Christoffersen, P.F. (1998). Evaluating interval forecasts. *International Economic Review*, 39(4), 841–862.

Czado, C. (2019). *Analyzing Dependent Data with Vine Copulas*. Springer, New York.

Embrechts, P., Klüppelberg, C., and Mikosch, T. (1997). *Modelling Extremal Events for Insurance and Finance*. Springer, Berlin.

Engle, R.F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007.

Engle, R.F., and Manganelli, S. (2004). CAViaR: Conditional autoregressive value at risk by regression quantiles. *Journal of Business and Economic Statistics*, 22(4), 367–381.

Fama, E.F. (1965). The behavior of stock-market prices. *Journal of Business*, 38(1), 34–105.

Fissler, T., and Ziegel, J.F. (2016). Higher order elicitability and Osband's principle. *Annals of Statistics*, 44(4), 1680–1707.

Genest, C., Rémillard, B., and Beaudoin, D. (2009). Goodness-of-fit tests for copulas: A review and a power study. *Insurance: Mathematics and Economics*, 44(2), 199–213.

Glosten, L.R., Jagannathan, R., and Runkle, D.E. (1993). On the relation between the expected value and the volatility of the nominal excess return on stocks. *Journal of Finance*, 48(5), 1779–1801.

Hill, B.M. (1975). A simple general approach to inference about the tail of a distribution. *Annals of Statistics*, 3(5), 1163–1174.

Kupiec, P.H. (1995). Techniques for verifying the accuracy of risk measurement models. *Journal of Derivatives*, 3(2), 73–84.

Mandelbrot, B. (1963). The variation of certain speculative prices. *Journal of Business*, 36(4), 394–419.

McNeil, A.J., and Frey, R. (2000). Estimation of tail-related risk measures for heteroscedastic financial time series: An extreme value approach. *Journal of Empirical Finance*, 7(3–4), 271–300.

Pickands, J. (1975). Statistical inference using extreme order statistics. *Annals of Statistics*, 3(1), 119–131.

Sklar, A. (1959). Fonctions de répartition à n dimensions et leurs marges. *Publications de l'Institut de Statistique de l'Université de Paris*, 8, 229–231.